Name: D S LAKSHMI NARAYANA

Register No.: 192512371

Course Name: SIGNALS AND SYSTEM

Course Code: ECA 0304

EXPERIMENT – 10

OBJECTIVE

• To distinguish the responses produced by initial stored energy and external excitation, obtain the impulse response of a continuous-time LTI system, and verify that the total response is the sum of the zero-input and zero-state responses, with the zero-state response also obtained through convolution.

Experimental Procedure

1. Define the continuous-time system dy(t)/dt + 2y(t) = x(t) and obtain H(s) under zero initial conditions.
2. Use MATLAB to display the transfer function and its pole.
3. Plot the impulse response and identify its decay pattern.
4. Obtain the zero-input response for y(0−) = 1 with x(t) = 0.
5. Obtain the zero-state response to x(t) = u(t) with zero initial conditions.
6. Compute the convolution of the unit-step input and impulse response, then compare it with the zero-state response.
7. Combine the zero-input and zero-state responses and verify the total response.

MATLAB CODING

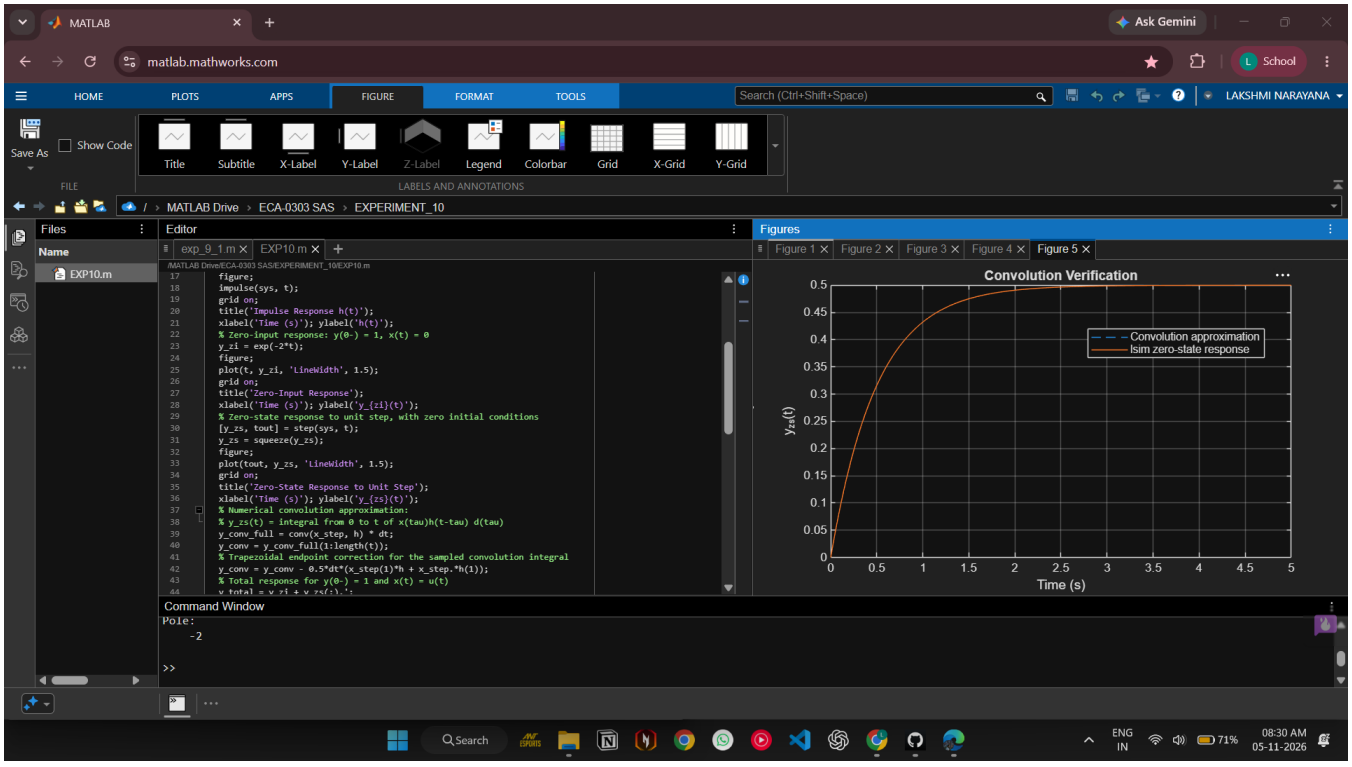
```
clc; clear; close all;
% Experiment 10: Zero-input, zero-state, impulse response and convolution
% System: dy(t)/dt + 2y(t) = x(t)
num = [1];
den = [1 2];
sys = tf(num, den);
disp('Transfer function:');
sys
disp('Pole:');
disp(pole(sys));
t = 0:0.01:5;
dt = t(2) - t(1);
```

```
h = exp(-2*t);                % Impulse response for t >= 0
x_step = ones(size(t));
% Impulse response
figure;
impulse(sys, t);
grid on;
% Unit-step input
title('Impulse Response h(t)');
xlabel('Time (s)'); ylabel('h(t)');
% Zero-input response: y(0-) = 1, x(t) = 0
y_zi = exp(-2*t);
figure;
plot(t, y_zi, 'LineWidth', 1.5);
grid on;
title('Zero-Input Response');
xlabel('Time (s)'); ylabel('y_{zi}(t)');
% Zero-state response to unit step, with zero initial conditions
[y_zs, tout] = step(sys, t);
y_zs = squeeze(y_zs);
figure;
plot(tout, y_zs, 'LineWidth', 1.5);
grid on;
title('Zero-State Response to Unit Step');
xlabel('Time (s)'); ylabel('y_{zs}(t)');
% Numerical convolution approximation:
% y_zs(t) = integral from 0 to t of x(tau)h(t-tau) d(tau)
y_conv_full = conv(x_step, h) * dt;
y_conv = y_conv_full(1:length(t));
% Trapezoidal endpoint correction for the sampled convolution integral
y_conv = y_conv - 0.5*dt*(x_step(1)*h + x_step.*h(1));
% Total response for y(0-) = 1 and x(t) = u(t)
y_total = y_zi + y_zs(:).';
figure;
plot(t, y_zi, t, y_zs(:).', t, y_total, 'LineWidth', 1.3);
```

```
grid on;
legend('Zero-input', 'Zero-state', 'Total response', 'Location', 'best');
title('Comparison of Response Components');
xlabel('Time (s)'); ylabel('Response');
% Compare convolution result with MATLAB lsim zero-state response
y_lsim = lsim(sys, x_step, t);
figure;
plot(t, y_conv, '--', t, y_lsim(:).', '-');
grid on;
legend('Convolution approximation', 'lsim zero-state response', ...
'Location', 'best');
title('Convolution Verification');
xlabel('Time (s)'); ylabel('y_{zs}(t)');
```

Input and Output Graphs

The MATLAB-generated input and output plots for the impulse, zero-input, zero-state, convolution and total-response tests are shown below.




MATLAB
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PLOTS
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FIGURE
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TOOLS
Search (Ctrl+Shift+Space)
LAKSHMI NARAYANA
Save As
Show Code
FILE
Title
Subtitle
X-Label
Y-Label
Z-Label
Legend
Colorbar
Grid
X-Grid
Y-Grid
LABELS AND ANNOTATIONS
MATLAB Drive
ECA-0303 SAS
EXPERIMENT_10
Editor
Figures
Figure 1
Figure 2
Figure 3
Figure 4
Figure 5
Convolution Verification
Convolution approximation
lsim zero-state response
Time (s)
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